

Let $H: M \rightarrow \mathbb{R}$. Then there is a unique vector field X_H with

$$i_{X_H} \omega := \omega(X_H, \cdot) = \Delta H.$$

Let φ_t be the family of diffeomorphisms generated by X_H . Then φ_t preserves

$$\omega: \quad \frac{d}{dt} \varphi_t^* \omega = \mathcal{L}_{X_H} \omega \stackrel{\text{Cartan}}{=} \underbrace{\Delta i_{X_H} \omega}_{\Delta H} + i_{X_H} \underbrace{\Delta \omega}_0 = 0.$$

The vector field X_H also preserves H :

$$X_H(H) = \Delta H(X_H) = i_{X_H} \omega(X_H, \cdot) = 0.$$

We call X **symplectic** if $i_X \omega$ is closed, **Hamiltonian** if it is exact. Note that Cartan's formula implies that X is symplectic iff $\mathcal{L}_X \omega = 0$.

For $f, g \in C^\infty(M, \mathbb{R})$, we define the **Poisson bracket**

$$\{f, g\} = \omega(X_f, X_g).$$

Prop: If X, Y are symplectic then $[X, Y]$ is Hamiltonian with Hamiltonian function $\omega(Y, X)$.

Proof:

$$\begin{aligned} i_{[X, Y]} \omega &= \mathcal{L}_X i_Y \omega - i_Y \mathcal{L}_X \omega, \\ &= \Delta i_X i_Y \omega + i_X \underbrace{\Delta i_Y \omega}_{\text{closed}} - i_Y \underbrace{\Delta i_X \omega}_{\text{closed}} - i_Y i_X \underbrace{\Delta \omega}_{=0}, \\ &= \Delta(\omega(Y, X)). \end{aligned}$$

□

Corollary: $X_{\{f, g\}} = -[X_f, X_g]$

Proof:

$$X_{\{f, g\}} = X_{\omega(X_f, X_g)} = [X_g, X_f].$$

□

Prop:

- $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0;$
- $\{f, gh\} = \{f, g\}h + g\{f, h\}.$

Prop: Let $f: M \rightarrow \mathbb{R}$. Then f is constant along integral curves of X_H iff $\{f, H\} = 0$.

Proof: $X_H(f) = i_{X_H} \Delta f = i_{X_H} i_{X_f} \omega = \omega(X_f, X_H) = \{f, H\}.$

□

Def: A Hamiltonian system (M, ω, H) is **integrable** if there are $n = \frac{1}{2} \dim M$ functions $f_1 = H, f_2, \dots, f_n$ such that

- $\{f_i, f_j\} = 0,$
- $\Delta f_1, \dots, \Delta f_n$ are lin. indep. at every point.